

Spatio-Temporal Reconnection for Multi-Robot Networks using Adaptive Prescribed-Time CBFs

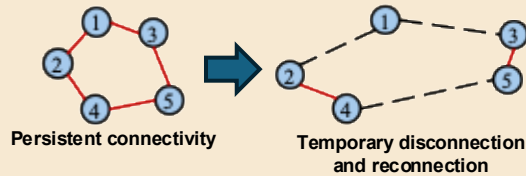
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Motivation

Multi-robot systems rely on communication links to support **coordination** and **information exchange**. Existing approaches often enforce **persistent global connectivity**, which may unnecessarily **limit robot motion** in large-scale task with limited communication range.



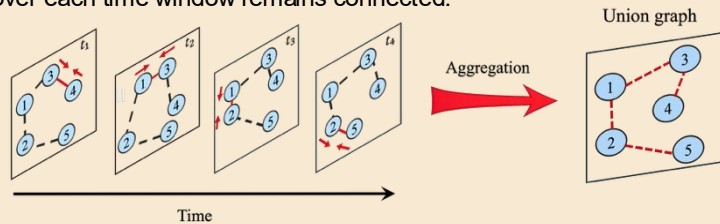
Challenges:

- Communication connectivity requirements constrain robot motion
- Predefining reconnection time may be infeasible
- Maintaining connectivity can interfere with task execution

Problem Formulation

Problem: Allow **temporary** disconnection across the time while guaranteeing periodic reconnection.

Formulation: We enforce joint global connectivity, the union graph over each time window remains connected.



How to adaptively determine when and which edge to connect using spatial and temporal information and finish reconnection without overly disrupt the task execution?

Method

A. Adaptive Prescribed-Time Control Barrier Functions

- **Prescribed-Time CBFs (PT-CBFs) [1]:** To reach the safe set within a **manually selected prescribed time**.
- **Finite-Time CBFs (FT-CBFs) [2]:** Reach the safe set in finite time with **state-dependent** convergence.
- **Adaptive PT-CBFs:** Compute the prescribed time **adaptively** from the initial state.

$$\dot{h}(x) + c\mu(t)\text{sign}(h(x))|h(x)|^\rho \geq 0$$

$$\mu(t) = \left(\frac{T_p}{T_p - t}\right)^m + \alpha, \quad t \in [0, T_p)$$

$$T_p = \frac{|h(x_0)|^{1-\rho}}{c(1-\rho)\mu_0}, \quad \mu_0 = 1 + \alpha > 1$$

B. Spatio-temporal Trigger Mechanism

Each communication edge must be reconnected within a time window $[t_k, t_{k+1}]$

Define the edge weight as

$$w_{ij}(t) = w_{\text{spatial}}(i, j) + w_{\text{temporal}}(t)$$

$w_{\text{spatial}}(i, j)$ measures the **effort** required to reconnect the edge

$w_{\text{temporal}}(t)$ measures how close the system is to the reconnection deadline

If $w_{i,j}(t) > 0 \rightarrow$ activate reconnection constraint between robot i and robot j

Solve QP :

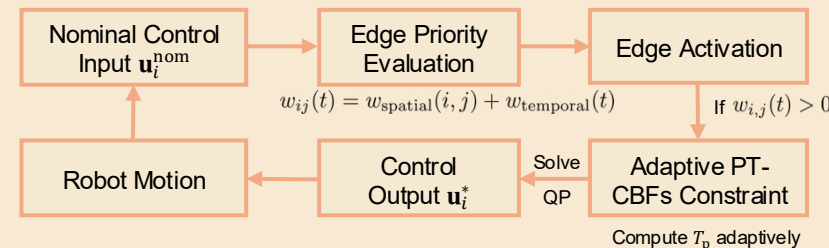
$$\mathbf{u}^*(t) = \arg \min_{\mathbf{u}} \sum_{i=1}^N \|\mathbf{u}_i - \mathbf{u}_i^{\text{nom}}(t)\|^2$$

s.t. collision avoidance constraints

adaptive PT-CBF reconnection constraints

control input bounds

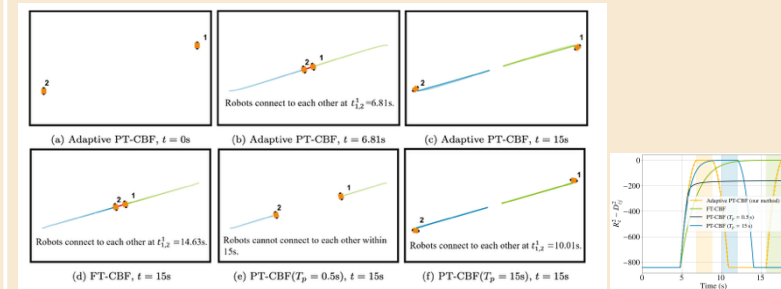
The controller computes a safe control input that satisfies safety and reconnection constraints.



Results

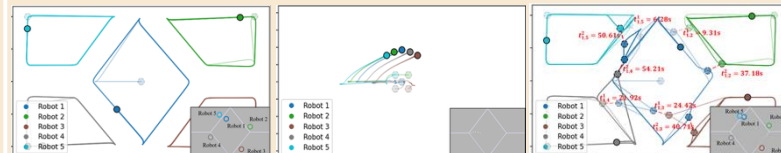
A. Two-robot-scenario Comparison

Two robots start outside the communication range and are initially stationary. The experiment compares different CBF formulations for communication reconnection.



Adaptive PT-CBF enables **feasible** and **less** conservative reconnection.

B. Multi-robot Case



Original Task

Persistent Connectivity

Our Adaptive PT-CBFs

- Task-only execution cannot ensure timely information exchange.
- Maintaining persistent connectivity overly restricts task execution.
- Our method achieves both **efficient** task execution and **timely** information exchange.

Conclusion

- **Adaptive PT-CBF** computes feasible reconnection time and enforce the constraint automatically
- **Spatio-temporal triggering mechanism** balances task execution and communication urgency
- Guarantees **reconnection within each time window**



[1] Abel, Imoleayo, et al. "Prescribed-time safety design for strict-feedback nonlinear systems." *IEEE Transactions on Automatic Control* 69.3 (2023): 1464-1479.

[2] Li, Anqi, et al. "Formally correct composition of coordinated behaviors using control barrier certificates." *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2018.